

Erratum: Theorem B of the (s, t) -square is Di Francesco–Kedem 1505.01657 Cor. 5.18

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Title: Erratum: Theorem B of the (s, t) -square is Di Francesco–Kedem 1505.01657
 Cor 5.18
Corrects: proofs/2026-10-02-st-square-closed.pdf, WIP content commit dd35498
 (sent to you earlier today).
WIP commit: This erratum’s content is work-in-progress commit 55f86c6
 (grandpa-rick/work-in-progress); this hash line was added in a follow-up commit.
Check script: scripts/day217dfk/dfk1505_check.py

The correction. The PDF at dd35498 presented Theorem B (the $t = 0$ edge) as new and asked whether you had seen it. It is not new. Both the operator and the identity

$$e_\mu^\star|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = \omega \tilde{H}_\mu(x; s) = \sum_\lambda \tilde{K}_{\lambda\mu}(s) s_{\lambda'},$$

$$E_k|_{t=0} = \sum_{|A|=k} \prod_{i \in A, j \notin A} \frac{x_i}{x_i - x_j} X_A T_{s,A},$$

are in P. Di Francesco and R. Kedem, *Difference equations for graded characters from quantum cluster algebra*, arXiv:1505.01657 (Transformation Groups 2018). The first equality is their Cor. 5.18. The second is a textbook identity. **That question is withdrawn.**

1. What DFK state (read first-hand)

Notation: $N = r + 1$ variables, $z_I = \prod_{i \in I} z_i$, $a_I(z) = \prod_{i \in I, j \notin I} \frac{z_i}{z_i - z_j}$ (eq. (1.1)), $\Gamma_I = \prod_{i \in I} \Gamma_i$, $\Gamma_i : z_i \mapsto qz_i$ ((5.2)–(5.3)).

(5.15) $M_{\alpha,n} = \sum_{I \subset [1,r+1], |I|=\alpha} (z_I)^n a_I(z) \Gamma_I$, once the condition $z_1 \cdots z_{r+1} = 1$ is dropped.

(5.16) $M_{\alpha,n} M_{\beta,p} = q^{\min(\alpha,\beta)(p-n)} M_{\beta,p} M_{\alpha,n}$. In particular the $M_{\alpha,1}$ commute.

(5.25) (Level 1 of Cor. 5.8, (5.18).) $\chi_n(q^{-1}, z) = q^{-\frac{1}{2} \sum n^{(\alpha)} \min(\alpha,\beta) n^{(\beta)} + \frac{1}{2} \sum \alpha n^{(\alpha)}} \prod_{\alpha=1}^r (M_{\alpha,1})^{n^{(\alpha)}} 1$.

(5.27) (Cor. 5.18.) “ $\chi_n(q^{-1}, z) = \lim_{t \rightarrow \infty} P_\lambda^{q,t}(z) = P_\lambda^{q^{-1},0}(z)$ ”, where $\lambda_i = n^{(i)} + \cdots + n^{(r)}$.

Rem. 5.20 also identifies the $M_{\alpha,1}$ with the Kirillov–Noumi raising operators in the limit $t \rightarrow \infty$.

2. Dictionary

DFK 1505.01657	ours (dd35498)
q	s
$z_1, \dots, z_{r+1}$	$x_1, \dots, x_N$
$M_{k,1}$	$E_k _{t=0}$ (same coefficient a_A , same shift $\Gamma_A = T_{s,A}$, same factor X_A)
$n^{(\alpha)}$	multiplicity of α as a part of μ
λ	μ'

Since $\sum_{\alpha,\beta} n^{(\alpha)} \min(\alpha,\beta) n^{(\beta)} = \sum_k \mu_k'^2$ and $\sum_\alpha \alpha n^{(\alpha)} = |\mu|$, the prefactor in (5.25) is $q^{-n(\mu)}$. So (5.25) and (5.27) say literally

$$\prod_i M_{\mu_i,1} \cdot 1 = q^{n(\mu)} P_{\mu'}(z; q^{-1}, 0) \quad (N = r + 1, \mu_1 \leq r),$$

which is the first equality of Theorem B. The bound $\mu_1 \leq r$ is removed by taking N large.

3. The bridge identity

Macdonald, *Symmetric Functions and Hall Polynomials*, Ch. VI (5.1) (our (M5)): with the plain involution ω ,

$$P_\lambda(x; q, 0) = \omega Q'_{\lambda'}(x; q).$$

With $\lambda = \mu'$ and $\tilde{H}_\mu(x; q) = q^{n(\mu)} Q'_\mu(x; q^{-1}) = \sum_\lambda \tilde{K}_{\lambda\mu}(q) s_\lambda$, this gives $q^{n(\mu)} P_{\mu'}(x; q^{-1}, 0) = \omega \tilde{H}_\mu(x; q)$, the second equality of Theorem B. The HL- Q versus Q' mismatch I suspected does not arise. The duality $\omega_{q,t} P_\lambda(q, t) = Q_{\lambda'}(t, q)$ uses the twisted $\omega_{q,t}$, not the plain ω .

4. Computer check

The script `scripts/day217dfk/dfk1505_check.py` works in $N = 4$ with symbolic q . For all 11 partitions μ with $|\mu| \leq 4$ it checks that the DFK product $\prod_i M_{\mu_i, 1} \cdot 1$ equals each of the following:

- (a) $q^{n(\mu)} P_{\mu'}(z; 1/q, 0)$, with P computed independently as a D_1 -eigenvector at generic (q, t) and then $t \rightarrow 0$;
- (b) $\sum_\lambda \tilde{K}_{\lambda\mu}(q) s_{\lambda'}$, from hard-coded Kostka–Foulkes polynomials;
- (c) our E_k -product at $t = 0$, implemented separately.

Result: **11/11 true** for (a), (b), (c) and (a)=(b). Example: $e_1 \star e_1|_{t=0} = q h_2 + e_2$.

Negative control: replace (a) by $P_{\mu'}(z; q, 0)$, without the inversion $q \mapsto 1/q$. This is false for 7 of the 11 partitions. The 4 that pass are the one-row $\mu = (n)$, where $P_{(1^n)} = e_n$ does not depend on q and $n(\mu) = 0$.

5. Updated novelty position

Item	Status
Theorem B ($t = 0$ edge)	Known. Cite DFK 1505.01657 (5.15), (5.25), Cor. 5.18 (5.27), together with Macdonald VI (5.1). Our contribution is only its placement as the $t = 0$ edge of the t -deformed $\star$, and a different derivation (from (N) plus triangularity).
Theorem A ($s = \infty$ edge $\rightarrow$ HL $P_{\mu'}(x; t)$)	Clean as checked. DFK 1505/1606/1704/2112/2303 contain no Hall–Littlewood, and their edge is always $t \rightarrow \infty$.
Lemma R (reflection)	Clean as checked.
Prop. C (collapse coefficient)	Clean as checked.
(N) ($\star$ is ∇ -transport)	Folklore-implicit (DFK 1704.00154; see also the 2112.09798 introduction).
Residual risk	Kirillov–Noumi 1999 (raising operators, including a possible $q \rightarrow \infty$ HL limit), and DFK arXiv:2112.09798 and 1908.00806. None of these has been read first-hand.

Revised question. Have you seen Theorem A anywhere? It states that $s^{-n(\mu)} e_\mu^\star \rightarrow P_{\mu'}(x; t)$ (Hall–Littlewood) as $s \rightarrow \infty$, i.e. the $q \rightarrow \infty$ limit at finite t of the t -deformed DFK/Kirillov–Noumi operators $M_{k;1}(q, t)$.